

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL2- ARTICLE REVIEW

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I _____ Venu G(192424390)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
- (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Selected Topic: Reinforcement Learning

Article Citation Details

Cabrejas-Egea, A., Zhang, R., & Walton, N. (2021). Reinforcement Learning for Traffic Signal Control: Comparison with Commercial Systems. *Transportation Research Procedia*, 58, 638–645. Elsevier.

DOI: 10.1016/j.trpro.2021.11.084

Domain: Intelligent Transportation Systems (ITS)

Application Area: Urban Traffic Signal Control

Machine Learning Problem: Reinforcement-learning-based adaptive traffic signal control.

Article DOI: <https://doi.org/10.1016/j.trpro.2021.11.084>

TASK 1: ARTICLE SUMMARY

(a) Domain and Specific ML Problem

The selected research article focuses on the application of Reinforcement Learning (RL) to intelligent traffic signal control. The application domain is Intelligent Transportation Systems, particularly the management of traffic at urban road intersections.

The specific machine-learning problem is to develop an intelligent traffic-signal controller that can automatically select appropriate signal phases according to the current traffic conditions. Conventional traffic signal systems generally use predefined timing plans or specialized traffic-control algorithms. These approaches may not respond optimally when traffic demand changes significantly.

The authors investigate whether Deep Reinforcement Learning (DRL) can learn effective traffic signal control policies by interacting with a simulated traffic environment. The RL methods are compared with established commercial traffic-control systems, including MOVA and Surtrac, as well as a cyclic traffic controller.

The study evaluates the controllers primarily using traffic delay and queue length. Therefore, instead of classification accuracy or F1-score, the important performance measures are cumulative delay and queue-related measures.

(b) Key Objective and Research Gap

The main objective of the article is to determine whether Deep Reinforcement Learning can provide effective traffic signal control and perform competitively against existing commercial traffic-control systems. The authors identified a research gap in earlier reinforcement-learning studies: many studies compared RL methods mainly with simple fixed-time traffic signal controllers rather than with sophisticated commercial adaptive systems. Consequently, it was difficult to determine whether the reported advantages of RL would remain when compared with more advanced real-world approaches. The authors attempted to address this gap by implementing several DRL approaches and evaluating them against commercial systems under comparable simulation conditions. They considered DDQN, DDDQN and A2C algorithms and used the PTV Vissim traffic simulator to create increasingly complex intersection scenarios. Traffic demand was varied to examine how well the learned policies generalized beyond their training conditions. Performance was evaluated using cumulative traffic delay and queue-related measures. The results showed that DRL, particularly DDDQN, could achieve lower delay and queue lengths than several conventional controllers in the tested scenarios.

TASK 2: METHODOLOGY ANALYSIS

(a) Machine Learning Techniques Used

The authors formulate traffic signal control as a Markov Decision Process (MDP).

An RL system consists of:

Agent: Traffic signal controller

Environment: Traffic intersection

State: Current traffic and signal information

Action: Selection of a traffic signal stage

Reward: Negative queue length

Objective: Maximize cumulative reward

The basic relationship can be represented as:

State → Agent → Action → Traffic Environment → Reward → Updated Policy

The principal algorithms investigated are:

1. Double Deep Q-Network (DDQN)

DDQN is an extension of Deep Q-Network (DQN). A neural network estimates the value of different actions. Double Q-learning helps reduce the overestimation problem associated with standard Q-learning.

2. Dueling Double Deep Q-Network (DDDQN)

DDDQN combines:

Deep Q-Network

Double Q-learning

Dueling network architecture

The dueling architecture separates the estimation of the value of a state from the advantage of individual actions.

3. Advantage Actor-Critic (A2C)

A2C combines two components:

Actor: Selects actions according to the learned policy.

Critic: Estimates the value of the current state.

The study found that the DQN-based methods performed strongly compared with A2C in the experiments.

Dataset and Experimental Environment

Unlike conventional supervised-learning studies, the article does not use a standard static dataset such as Iris or MNIST.

Instead, the authors generated traffic data using the PTV Vissim traffic simulator.

Parameter

Description

Data source

Vissim simulation

Application

Traffic signal control

State information

lengths and current signal stage

Reward

Negative sum of queue lengths

Training demand

vehicles/hour per incoming lane

PTV

Queue

400

DQN training episodes	400
A2C training episodes	100
Evaluation	
One-hour traffic simulations	
Main metrics	
Cumulative delay and queue length	

The traffic environment was prepared using different intersection configurations and traffic-demand levels. The increasing complexity of the scenarios allowed the authors to investigate whether the RL agents could maintain their performance when the traffic-control problem became more difficult.

Experimental Scenarios

Experiment 1 – Cross Straight

The first scenario represents a relatively simple four-arm intersection with two signal stages and no turning movements.

Experiment 2 – Cross Triple, 4 Actions

The second scenario increases the complexity by introducing three lanes on incoming links and allowing turning movements.

Experiment 3 – Cross Triple, 8 Actions

The third experiment provides a larger action space with eight possible signal stages, making the signal-control problem more complex.

(b) Mapping to CO4 Topics

CO4 Topic

Relation to Article

Inductive Learning agent learns a useful traffic-control policy from repeated interaction with the environment.	The
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Decision Trees directly used in the research.	Not
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Statistical Learning Neural networks learn value estimates from observed states and rewards.	
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Reinforcement Learning topic of the article. The agent learns through rewards and penalties.	Main
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Deep Learning
and DDDQN use neural networks for value-function approximation.

DQN

Markov Decision Process
Traffic signal control is represented using states, actions, rewards and transitions.

Methodological Strength

A major strength is that the researchers compare reinforcement-learning methods against commercial traffic-control systems, rather than only against a simple fixed-time controller. This makes the experimental comparison more meaningful.

Methodological Limitation

The main limitation is that the evaluation is simulation-based. Although PTV Vissim provides a realistic traffic simulation environment, it cannot completely represent real-world traffic conditions. Real intersections may contain pedestrians, accidents, emergency vehicles, sensor errors, unusual traffic patterns and weather-related changes.

TASK 3: RESULTS AND CRITICAL EVALUATION

(a) Key Results

Since this is a reinforcement-learning control problem, conventional classification measures such as accuracy, precision, recall and F1-score are not appropriate. The main performance measures are:

Cumulative delay

Average queue length

Maximum queue length

Experiment 1 Results

Controller

Cumulative Delay (seconds)

Average Sum of Queues (m)

Cyclic

143,660.50

132.37

MOVA

27,187.53

60.59

SURTRAC

29,008.36

72.41

A2C

26,382.14

56.07

DDQN

28,303.94

50.11

DDDQN
21,286.86

49.42

DDDQN produced the lowest cumulative delay and average queue among the listed controllers in this experiment.

Experiment 2 Results

Controller Cumulative Delay (seconds)
Average Maximum Queue (m)

MOVA
260,257.65
179.27

MOVA*
135,220.91
153.58

DDDQN
155,563.22
128.20

The results indicate that the RL controller was able to reduce queue-related performance measures compared with the standard MOVA configuration.

Experiment 3 Results

Controller
Cumulative Delay (seconds)
Average Maximum Queue (m)

MOVA
165,456.44
339.41

MOVA*
72,642.59
123.52

DDDQN
71,245.61
119.86

The DDDQN controller produced slightly lower cumulative delay and average maximum queue than the MOVA* configuration in this experiment.

(b) Critical Evaluation

Are the Results Realistic?

The results are promising because the researchers used a commercial traffic simulator rather than an extremely simplified artificial environment. The RL agents also demonstrated performance under different traffic-demand levels.

However, the results should not be interpreted as proof that RL will always outperform commercial traffic-control systems in real-world deployments.

The experiments are still based on simulation. Real-world intersections can have:

Pedestrians

Emergency vehicles

Road accidents

Weather changes

Sensor failures

Road construction

Unpredictable driver behaviour

Communication failures

Therefore, real-world validation is necessary.

Potential Biases

1. Simulation Bias

The results depend on how accurately PTV Vissim represents real traffic.

2. Training Bias

The RL agents were trained using particular traffic-demand conditions. Different cities and intersections may have significantly different traffic distributions.

3. Limited Number of Scenarios

Only a limited number of intersection configurations were evaluated.

4. Baseline Configuration

Commercial traffic-control systems may require careful site-specific tuning. Differences in configuration could influence the comparison.

5. Reproducibility

RL algorithms can have variability between training runs. Multiple random seeds and statistical analysis would provide stronger evidence.

Additional Experiments Recommended

The research could be strengthened by:

Testing on real-world traffic datasets.

Evaluating more intersections and different cities.

Running experiments with multiple random seeds.

Reporting mean, standard deviation and confidence intervals.

Testing accidents and emergency-vehicle scenarios.

Comparing against modern algorithms such as PPO, SAC and MAPPO.

Testing the system using real traffic signals.

Measuring fuel consumption and CO₂ emissions.

Evaluating pedestrian waiting time.

Performing long-term real-world pilot testing.

(c) Real-World Applicability

The proposed RL approach can potentially be used in:

Smart cities

Intelligent Transportation Systems

Adaptive traffic signals

Urban intersections

Highway junctions

Traffic management centres

Connected-vehicle systems

An intelligent traffic controller could receive information from cameras, loop detectors, radar sensors or connected vehicles and select an appropriate signal phase.

Industry Challenges

Data Availability

The system needs reliable real-time traffic information.

Computational Requirements

Training deep RL models can require significant computational resources.

Safety

Traffic signals are safety-critical systems. An incorrect action could cause dangerous traffic conditions.

Generalization

An agent trained at one intersection may not perform equally well at another intersection.

Explainability

Traffic authorities may require explanations for decisions made by an AI controller.

Regulation

AI-based traffic-control systems must comply with transportation and safety regulations.

TASK 4: REFLECTION AND LEARNING OUTCOME

(a) Three Key Insights

Insight 1: Reinforcement Learning Learns Through Interaction

The most important insight is that reinforcement learning does not require a predefined correct answer for every input. Instead, the agent interacts with an environment and learns which actions produce better rewards.

CO4 Connection: Reinforcement Learning.

In this article, the traffic signal controller learns from the traffic environment and receives rewards based on queue conditions.

Insight 2: Reward Design Is Extremely Important

The article helped me understand that the reward function directly influences the behaviour of an RL agent.

The authors use negative queue length as the reward. Therefore:

Smaller queue → Better reward

Larger queue → Worse reward

CO4 Connection: Reinforcement Learning and Statistical Learning.

A poorly designed reward function could cause an RL system to optimize the wrong objective.

Insight 3: Algorithm Architecture Influences Performance

The research demonstrates that different RL architectures can produce different performance. DQN-based approaches, particularly DDDQN, performed strongly compared with A2C in the reported experiments.

CO4 Connection: Deep Learning and Reinforcement Learning.

This shows that performance depends not only on the learning framework but also on the network architecture, state representation, reward function and action space.

(b) Proposed Novel Experiment

Multi-Agent PPO/MAPPO for Coordinated Traffic Signals

If I were to extend this research, I would develop a Multi-Agent Proximal Policy Optimization (MAPPO) system for controlling multiple interconnected traffic intersections.

In the original study, traffic signal control is primarily evaluated at individual intersections. A more advanced system could allow neighbouring intersections to communicate.

Each intersection would act as an RL agent and share information such as:

Queue length

Traffic density

Average vehicle speed
Current signal phase
Incoming vehicle flow
Downstream congestion

The agents could jointly learn a policy that minimizes network-wide traffic congestion.

Feasibility

The proposed experiment is feasible because the original study already demonstrates the use of RL with a traffic simulator. The same simulation environment could be extended to multiple connected intersections.

Expected Impact

The proposed MAPPO system could:

Reduce network-wide congestion.
Improve coordination between traffic signals.
Reduce vehicle waiting time.
Improve traffic flow.
Handle larger urban road networks.
Provide a stronger comparison between traditional RL and multi-agent RL.

A useful experimental comparison would be:

MOVA vs DDDQN vs PPO vs MAPPO

using the same traffic network and traffic-demand conditions.

CONCLUSION

The article Reinforcement Learning for Traffic Signal Control: Comparison with Commercial Systems provides an important application of machine learning to intelligent transportation. The researchers demonstrate that Deep Reinforcement Learning can learn traffic signal policies and achieve competitive performance against established traffic-control systems.

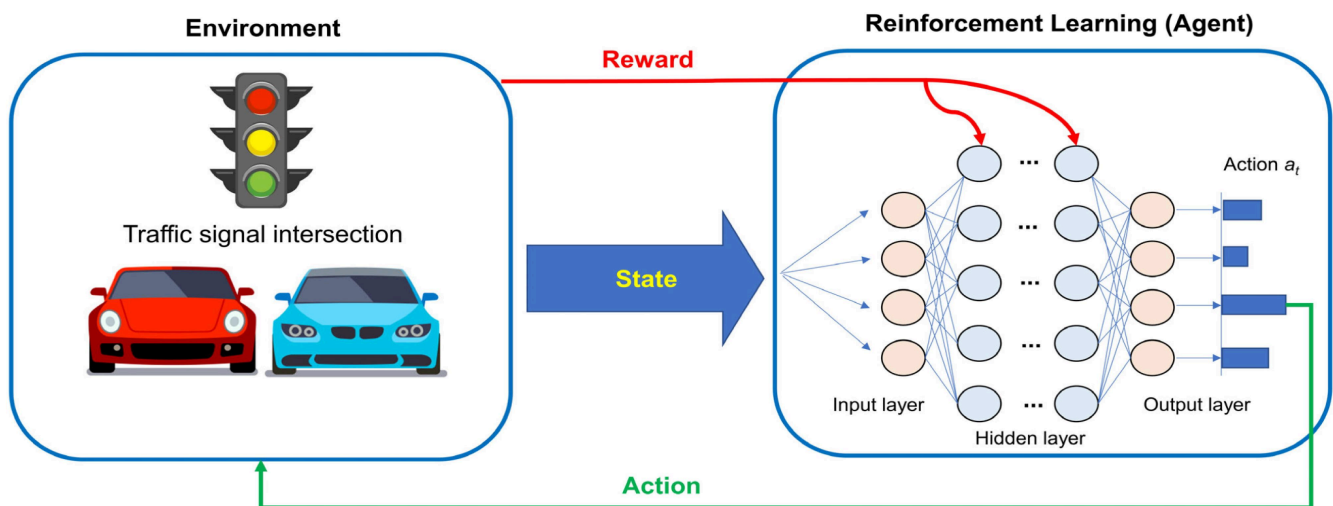
Among the investigated approaches, DDDQN showed particularly strong performance in the reported experiments. The study is valuable because it does not restrict the comparison to simple fixed-time traffic signals and instead considers commercial systems such as MOVA and Surtrac.

However, the research is primarily simulation-based. Therefore, further research is required before deploying the approach in real-world traffic systems. Real-world datasets, multiple intersections, safety testing, statistical validation and modern multi-agent RL methods could strengthen the findings.

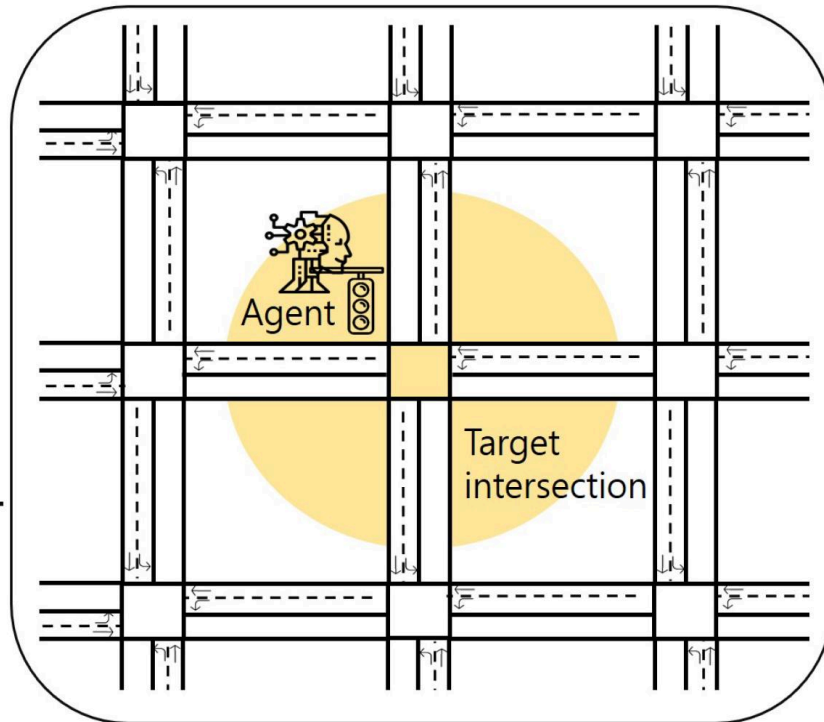
Overall, the article demonstrates how Reinforcement Learning, Deep Learning and Markov Decision Processes can be combined to solve practical traffic-control problems. It provides a strong example of the application of CO4 concepts to a real-world engineering problem.

REFERENCES

1. Cabrejas-Egea, A., Zhang, R., & Walton, N. (2021). Reinforcement Learning for Traffic Signal Control: Comparison with Commercial Systems. *Transportation Research Procedia*, 58, 638–645. Elsevier. DOI: 10.1016/j.trpro.2021.11.084.
2. Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction*. MIT Press.



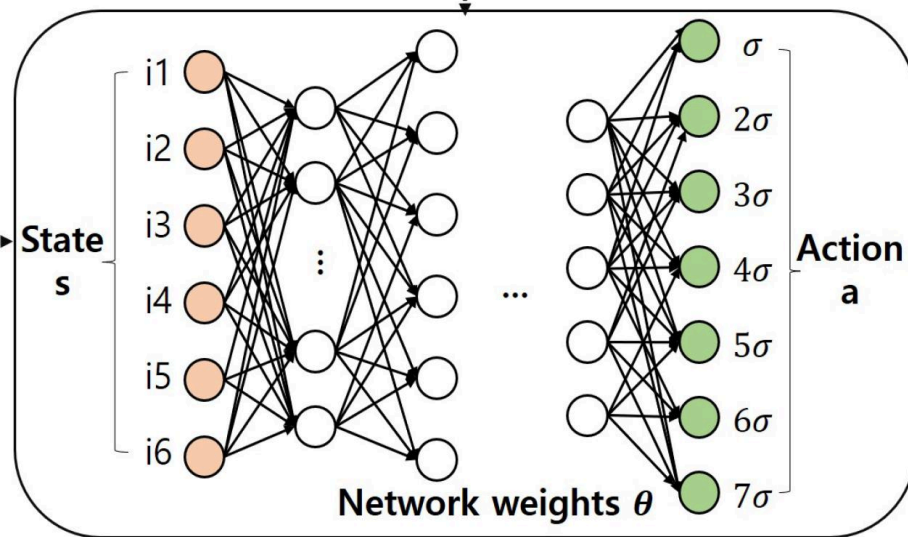
Environment



Perceived State

Reward

Sent Action



DQNAgent

